

PS 647 — GROUP PROJECT

Digital Financial Infrastructure and Financial Inclusion in India

A state-level panel study of the access–usage gap

Author One

Roll No. XXXXXXXX

Author Two

Roll No. XXXXXXXX

Author Three

Roll No. XXXXXXXX

Department of [Department Name]
[Institution Name]

Course instructor: [Instructor Name]

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Abstract

India presents a paradox. Adult account ownership rose from 35 per cent in 2011 to roughly 89 per cent by 2025, yet India records the largest share of *inactive* accounts in the world: about 35 per cent of Indian adults who hold an account hold a dormant one, some seven times the average across other developing economies (Demirgüç-Kunt et al., 2022; World Bank, 2025). Accounts have been opened far faster than they have been used. This report asks whether the expansion of financial infrastructure is itself part of that story — whether expansion has been formal rather than functional.

We build verified state-level panels from nine sources: eight Reserve Bank of India and Government of India series covering branches, ATMs, deposit and credit account counts, deposit and credit balances, per-capita state income and adult population, plus state-level digital payment data from PhonePe Pulse. All-India aggregates reconcile to published figures to within 0.001 per cent. Hypothesis 1 — that infrastructure raises *Access* more than *Usage* — is tested on three tracks by two-way fixed-effects regressions of *Access*, *Usage* and their difference, with standard errors clustered by state. The decision rule for accepting or rejecting it was fixed in writing before any coefficient was estimated.

On the two **physical** tracks — 33 states over FY2018–FY2023 and 32 over FY2012–FY2023 — **the hypothesis is not supported, and the null is uninformative**. A diagnostic run before estimation, with thresholds fixed in advance, found that only 7.0 per cent of the variation in branches per lakh adults and 8.7 per cent of that in ATMs occurs *within* states; roughly 93 per cent is cross-sectional, which is exactly what fixed effects discard. A Hausman test then rejects random effects decisively, ruling out the estimator that would recover that variation. The fragility this implies is demonstrated rather than asserted: dropping two observations out of 198 reverses the sign of every ATM coefficient.

On the **digital** track — 33 states over FY2019–FY2023, adding UPI adoption — the identification problem disappears. Within-state variation in UPI users per adult is 67.1 per cent, and explanatory power rises from a within- R^2 of 0.069 to 0.609 for *Access*. UPI adoption raises *Access* (+0.439) more than *Usage* (+0.183), and the gap is significant ($p = 0.020$) — the first time in this study the access–usage gap is distinguishable from zero. Decomposing the outcome shows the effect runs *entirely through credit accounts, not deposit accounts*, which is the reverse of what an accounting artefact would produce, since deposit accounts are the component definitionally entangled with UPI registration. The result survives dropping any single state.

Two caveats bound this. The pre-committed hypothesis named branches and ATMs and it failed; UPI was added *after* seeing that null, so the digital result is exploratory by construction — a hypothesis generated by this data, not one tested and passed. And while the identification problem is solved, the causal one is not: fixed effects do not remove state-specific time-varying confounders.

Keywords: financial inclusion; access–usage gap; banking infrastructure; digital payments; UPI; panel data; two-way fixed effects; India; pre-registration.

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Chapter 1

Introduction, Background and Literature

1.1 The puzzle

Between 2011 and 2017 the share of Indian adults holding a bank account rose from 35 to 80 per cent (Demirgüç-Kunt et al., 2022), reaching roughly 89 per cent by the 2025 Global Findex round (World Bank, 2025). Much of that movement came from a single programme: the Pradhan Mantri Jan Dhan Yojana (PMJDY), launched in August 2014, under which more than 56 crore accounts had been opened by August 2025 (Press Information Bureau, Government of India, 2025). Measured by the indicator that dominates international league tables, India solved financial exclusion inside a decade.

The same surveys record a second fact that sits uncomfortably beside the first. India has the largest share of *inactive* accounts of any country measured: roughly 35 per cent of Indian adults who own an account report that it is dormant, some seven times the 5 per cent average across other developing economies (Demirgüç-Kunt et al., 2022). Official figures point the same way — about 26 per cent of PMJDY accounts, some 15 crore, are reported inoperative, with an average balance of roughly Rs. 4,808 (Press Information Bureau, Government of India, 2025). Among adults holding a dormant account, 49 per cent cite distance to the institution and 46 per cent say they did not need an account (Demirgüç-Kunt et al., 2022).

An account that exists but is never used counts as inclusion in every headline statistic while delivering none of the consumption-smoothing, risk-pooling, or credit benefits that justify the policy effort. The gap between *having* an account and *using* one is the central empirical object of this report, and we refer to it throughout as the **access–usage gap**.

The gap could arise from low incomes, informality, distrust, poor financial literacy, or the simple fact that many accounts were opened to receive a one-off transfer. This report examines a candidate that policy can actually move: the infrastructure through which banking is delivered. The reasoning is asymmetric in an important way. Opening an account is largely a *supply-side* decision — banks can be given targets and audited against them, and Indian branch licensing has been used this way since the 1970s. Using an account is a *demand-side* decision shaped by income, informality, and trust. Infrastructure relaxes the first constraint directly; whether it relaxes the second is an empirical question. If the asymmetry is real, expanding infrastructure should raise measured access more than measured usage, and the gap should *widen* as infrastructure expands. Expansion would be, in a phrase used throughout, **formal rather than functional**.

1.2 Background

Three policy waves have shaped Indian financial inclusion, and this study’s sample period sits inside the third. Following bank nationalisation in 1969, the RBI operated a licensing rule under which a bank could open a branch in an already-banked location only if it also opened four in unbanked ones (Burgess and Pande, 2005); the resulting 1977–1990 expansion remains the most studied episode in the Indian financial-development literature. From the mid-2000s financial inclusion became an explicit objective, bringing the no-frills account, the business correspondent model, and the Rangarajan Committee report (Committee on Financial Inclusion, 2008), together with measurement frameworks (Sarma, 2008; Sarma and Pais, 2012). From 2014, PMJDY drove ownership toward saturation while the technology stack sometimes called “India Stack” — biometric identity, interoperable real-time payments, and consented data sharing — rewired how retail payments are made (D’Silva et al., 2019). The Unified Payments Interface processed roughly 83.7 billion transactions in calendar 2023, 131.1 billion in 2024, and 228.3 billion in 2025 (National Payments Corporation of India, 2025). This report’s sample window sits almost exactly on top of that transition, a point taken up in Section 3.5.

India also measures inclusion officially. The RBI’s composite Financial Inclusion Index rose from 64.2 in March 2024 to 67.0 in March 2025 (Reserve Bank of India, 2025a), and is built from three parameters with fixed weights: **Access 35 per cent, Usage 45 per cent, Quality 20 per cent**. Two observations follow. First, the RBI’s own scheme weights usage *above* access — an implicit acknowledgement that account counts overstate inclusion. Second, those weights exist to collapse three dimensions into one number, whereas this report does the opposite: it holds Access and Usage apart precisely so their *difference* can be estimated and tested. The weights are therefore not applicable here, and Section 3.2 records why applying them would have been an error rather than a refinement.

1.3 Literature

Access as a supply-side lever. The foundational result is Burgess and Pande (2005), who exploit the 1977–1990 licensing rule as policy-driven variation in rural branch openings and find that branch expansion reduced rural poverty. The paper established both a substantive claim — that physical banking infrastructure has real welfare consequences — and a methodological template: administrative state panels, with infrastructure variation that is not simply a response to local prosperity. The broader literature is consistent: Beck et al. (2007) find financial development disproportionately raises the incomes of the poor, and Allen et al. (2016) document that cost and proximity are among the strongest correlates of both ownership and use. One caution prefigures our own conclusion. Buliskeria and Baxa (2022) revisit Burgess and Pande (2005) and find the poverty effect does not survive once additional policy breaks are accounted for. The lesson is not that branches do not matter, but that in this literature the identifying variation is often thinner and more fragile than it appears, and claims should be stress-tested against perturbations of the sample. Chapter 3 does exactly that, and the stress test produces its most informative result.

Access is not usage. A large experimental literature establishes that removing the barriers to *opening* an account does not reliably produce *use* of it. The clearest statement is Dupas et al. (2018), who randomly offered free basic accounts in Uganda, Malawi, and Chile: over two years only 17, 10, and 3 per cent of treated individuals respectively made five or more deposits, and intention-to-treat effects on savings were indistinguishable from zero. Their conclusion is the proposition this report takes to Indian administrative data — “policies merely focused on expanding access to basic accounts are unlikely to improve welfare noticeably on

average.” Karlan et al. (2014) reach a compatible position, finding the binding constraints frequently behavioural and transactional rather than a matter of account availability. This report’s contribution to the strand is one of scale and setting: the experimental evidence is individual-level and drawn from three countries, and we ask whether the same wedge is visible in the administrative aggregates of a country of 1.4 billion people, and whether it responds to the infrastructure variable Indian policy actually controls.

Measuring inclusion. How inclusion is measured determines what can be found. Sarma (2008); Sarma and Pais (2012) construct a multidimensional index combining penetration, availability, and usage; the RBI’s FI-Index follows the same logic. Composite indices suit ranking and monitoring, but they are poorly suited to the question asked here: if access and usage are averaged into one index, the wedge between them becomes unobservable by construction — and the wedge is the object of interest. This report therefore departs from the index tradition deliberately.

Digital payments and the substitution of channels. The final strand asks whether physical infrastructure is still the relevant infrastructure. Suri and Jack (2016) show that expanded mobile-money access in Kenya produced long-run poverty reductions operating through risk-sharing and occupational change rather than account ownership as such. Crouzet et al. (2023) show how India’s 2016 demonetisation triggered durable adoption of electronic payments, shaped by network effects rather than infrastructure stock. D’Silva et al. (2019) argue that the design of India’s digital public infrastructure, rather than the density of physical bank premises, is what changed the cost of a transaction. These matter for interpretation: if the marginal channel for financial activity shifted from the branch to the phone during our window, a null result for branches and ATMs may be a substantive statement about channel substitution rather than a measurement failure.

Where this report sits. The experimental literature establishes the access–usage wedge at the individual level but cannot speak to national infrastructure policy; the Indian panel literature establishes that infrastructure has real effects but treats inclusion as a single dimension. We combine the two — an administrative state panel, with access and usage held apart as separate outcomes — so that the *differential* effect of infrastructure can be estimated directly.

1.4 Hypotheses

The report tests three hypotheses. The first is developed and estimated in full in Chapter 3; the remaining two are reported in Chapters 4 and 5.

Hypothesis 1 (Chapter 3)

Greater financial infrastructure raises financial inclusion, but its effect on **access** is stronger than its effect on **usage**.

Rationale. Opening an account is a supply-side decision that banks can be pushed to make. Using an account is a demand-side decision shaped by income, trust, and informality. Infrastructure removes the first constraint, not the second.

Expected: $\beta_{\text{access}} > \beta_{\text{usage}} > 0$, with the difference statistically significant.

To be completed: State Hypotheses 2 and 3 in the same format — claim, rationale, expected sign pattern — then develop them in Chapters 4 and 5.

Hypothesis 2 (Chapter 4)

[Statement to be supplied.]

Hypothesis 3 (Chapter 5)

[Statement to be supplied.]

1.5 Contribution and roadmap

This report makes four contributions. **First**, a verified state-year panel assembled from nine sources — eight official series using three different year conventions, three different state-boundary vintages and two different bank universes, plus state-level digital payment data — whose all-India aggregates reproduce published figures to within 0.001 per cent, with no imputation and no missing cells. **Second**, a direct test of the access–usage wedge on administrative data, in which the wedge is estimated as its own equation and therefore carries its own standard error rather than being inferred by eye from two coefficients. **Third**, a decision rule written down with its consequences before any coefficient was computed and never revised, which proved load-bearing twice over: it delivers the verdict on physical infrastructure, and it is what forces the honest statement that the digital result — added after that verdict was known — is exploratory rather than a test that was passed. **Fourth**, a negative result that turns into a positive one by diagnosis rather than by search. Physical infrastructure stocks move too slowly within states for a short fixed-effects panel to identify their effect; the same diagnostic that establishes this identifies digital payment adoption as a measure that does move, and on it the predicted pattern appears — running entirely through credit accounts rather than deposit accounts, which is the reverse of what a mechanical artefact would produce.

Chapter 2 documents the panels. Chapter 3 develops and tests Hypothesis 1 in full, on three tracks: two measuring infrastructure physically, and one substituting digital payment adoption. Chapters 4 and 5 present the second and third hypotheses, and Chapter 6 synthesises across the three and concludes. Appendix A reproduces the full decision log.

Chapter 2

Data and Panel Construction

This chapter documents the panel the hypothesis chapters draw on. It comes first because the construction is shared, and because several choices made here — the year-alignment rule above all — silently corrupt results if got wrong and are invisible in the output if got right. Every choice was logged when it was taken, with its rationale; the log is Appendix A, and decisions are cited below by identifier (D1, D2, ...).

2.1 Unit of analysis and coverage

The unit of observation is the **state–financial year**, and every variable is a **stock as at 31 March**, not a flow. A financial year is identified by the calendar year of the 31 March on which it closes: `fy_end = 2018` denotes the position as at 31 March 2018, the close of financial year 2017–18.

Table 2.1: Panel coverage

Panel	Window	Years	States	N	Infrastructure measured
Track 1	FY2018–FY2023	6	33	198	Branches and ATMs
Track 2	FY2012–FY2023	12	32	384	Branches only
Track 3	FY2019–FY2023	5	33	165	Branches, ATMs and UPI

All three panels are **balanced and complete**: every state appears in every year, with no missing cells and no non-positive values in any series. The state counts differ for one documented reason given in Section 2.4. ATMs cannot be extended before 2018, because RBI’s state-wise release begins there, which is why Track 2 carries branches alone; Track 3 starts at FY2019 because that is the first complete financial year of digital payment data. Track 3’s banking columns are asserted **identical** to Track 1’s, so the earlier results remain reproducible from an untouched file and the tracks stay directly comparable.

2.2 Sources

Nine series are used. Eight are official publications of the RBI or the Government of India, with one academic redistribution filling a single year; the ninth is the digital payment series introduced for Track 3.

Table 2.2: Source of each series

Series	Publication	Table / report	Role
Bank branches	RBI <i>Handbook of Statistics on Indian States</i> , 2024–25	Table 152, bank offices of scheduled commercial banks, end-March	Independent
ATMs	RBI <i>State-wise and Region-wise Deployment of ATMs</i>	Standalone quarterly release; end-March files 2018–2024	Independent
Deposit accounts	RBI Basic Statistical Returns (BSR-2), via DBIE/CIMS	Report 853, Annual BSR-2 Table 2.3; 31 March 2018 from India Data Portal	Access
Credit accounts	RBI Basic Statistical Returns (BSR-1), via India Data Portal	Derived state-year credit accounts 2010–2023, including RRBs	Access
Deposits outstanding	RBI <i>Handbook</i>	Table 155, deposits of SCBs (Rs. crore)	Usage
Credit outstanding	RBI <i>Handbook</i>	Table 156, credit of SCBs (Rs. crore)	Usage
Per-capita NSDP	RBI <i>Handbook</i>	Table 19, per-capita NSDP, current prices	Control
Adult population	Technical Group on Population Projections (2020)	Annual totals with quinquennial age structure; 1 March series	Denominator
Digital payments	PhonePe Pulse open data release	Registered users, transaction count and value, by state and quarter	Independent

Retrieval was non-trivial in each case. The Handbook tables ([Reserve Bank of India, 2025b](#)) are HTML only, each split across two blocks covering earlier and later years, and were parsed and stitched programmatically. The ATM release ([Reserve Bank of India, 2024a](#)) is navigable only through an ASP.NET postback widget rather than plain URLs, so per-release permalinks were harvested by scripting those postbacks. The BSR returns ([Reserve Bank of India, 2024b](#)) are no longer published as static files at all — the legacy links redirect to a single-page application — and were retrieved programmatically from the DBIE/CIMS API; the 31 March 2018 deposit-account figures, which the current report does not reach back to, come from an academic redistribution of the same returns ([India Data Portal, 2024](#)). Adult population derives from the official projections report ([Technical Group on Population Projections, 2020](#)).

2.3 The year-alignment rule

Three year conventions are in play, and they disagree on labels even when they agree on dates. This is the easiest place in the project to introduce a silent one-year offset — an error that produces no missing value, no warning, and no implausible number, and simply biases every estimate. The Handbook banking tables and the ATM release label a stock as at 31 March 2018 as 2018; the Handbook NSDP tables call the same financial year 2017–18; the population projections date their stock 1 March 2018.

Alignment rule (D5, D9)

Banking year $Y \equiv$ NSDP financial year $(Y - 1) - Y \equiv$ population at 1 March Y .

All three are recorded in one canonical column, **fy_end**, added to all eight input files. Merging and filtering use **fy_end** only, and it supplies the year fixed effect.

Of the eight files only the income file required conversion; native labels were retained alongside **fy_end** so the original coding stays auditable. The 1 March population series is used rather than 1 July or 1 October: every numerator is a stock at 31 March, so the denominator must be a stock at approximately the same instant — one month away rather than six. The rule was verified against published figures rather than assumed, as Section 2.5 records.

2.4 State harmonisation

Fixed effects require each unit to denote the same thing in every year. India’s boundaries changed inside both windows and, more awkwardly, different sources record the same change in different years. Three adjustments follow.

Units dropped (D1). Lakshadweep and the combined Dadra & Nagar Haveli and Daman & Diu are dropped: no per-capita NSDP is published for either in any year, so neither can carry the income control. Both are very small; the panel goes from 35 units to 33.

Units merged (D2). Jammu & Kashmir with Ladakh, and Dadra & Nagar Haveli with Daman and Diu, *in every year*. Merging only from the year of the formal change fails because the sources disagree about when that was: in the Handbook, Daman & Diu goes blank from 2020 exactly as Dadra & Nagar Haveli jumps from 59 to 110 branches, whereas the ATM files keep three separate columns until 2021. Summing all components in all years is the only rule producing a continuous series in both sources simultaneously.

Andhra Pradesh and Telangana, twelve-year panel only (D13). Telangana was carved out of Andhra Pradesh in June 2014, so in the twelve-year panel it shows *zero* branches, deposits and credit for 2012–2014, its activity being recorded inside Andhra Pradesh. Branches confirm the mechanism exactly: Andhra Pradesh falls from 9,900 in 2014 to 6,290 in 2015 as Telangana appears with 4,721, and the combined series runs continuously from 9,900 to 11,011. The two are therefore summed in every year, which is why Track 2 has 32 units. The failure mode is instructive: had the merge been missed, three state-years would have entered recording zero infrastructure alongside zero deposits — outliers positioned precisely to manufacture a spurious *positive* association between infrastructure and inclusion, the sign the hypothesis predicts. It was caught by a completeness check flagging zeros, not by inspecting results.

General rule

Merges are performed on **raw levels** — branches, ATMs, deposits, credit, account counts, population — **before** any ratio is computed. Ratios are never averaged across merged units.

After upper-casing and standardising “AND” to “&”, 34 of 37 units match exactly across sources; the exceptions are Delhi, Jammu & Kashmir, and Daman & Diu, each named differently in the population or ATM files. Matching uses **word boundaries** rather than substrings: a naive substring test folds ANDAMAN into the Daman group, an error that occurred once during preparation and is now guarded by an explicit assertion.

2.5 Validation

The panel was checked against independently published aggregates rather than accepted on trust. All-India ATM totals recomputed from the state columns match RBI’s published national figures exactly in every year 2018–2024 (222,066 in 2018 through 257,654 in 2024; the 2024 decline is genuine decommissioning, not a parsing error). State-by-year deposit amounts differ from Handbook Table 155 by a median of 0.0001 per cent, and credit amounts from Table 156 by 0.001 per cent. Population totals reproduce from the sum of age bands, and the published 18+ figure reproduces from a residual identity, for all 23 states that print it. On year alignment, Maharashtra at `fy_end = 2018` returns 12,545 branches, 25,651 ATMs and per-capita income of Rs. 172,663 — all matching RBI’s published values exactly, confirming no offset was introduced. Completeness is asserted in code: 198 and 384 rows, balanced, no nulls, no zeros.

2.6 The digital payment series

Track 3 requires a measure of digital financial infrastructure at state level. India has no official state-wise UPI release covering the full window, so the series used is PhonePe Pulse (PhonePe, 2026), the open data release of one of the two largest UPI applications: registered users, transaction counts and transaction value, by state and calendar quarter.

Conversion to the panel’s year convention follows the same rule as everything else. Transactions and value are **flows**, summed over the four quarters making up each financial year; registered users is a **stock**, taken at the quarter ending 31 March and verified monotone across all quarter transitions. Denominators follow the same access/usage distinction as the banking series: registered users **per adult**, transactions and value **per capita**. Coverage is complete — 36 raw states across 34 quarters with no gaps and no missing amounts — and every one of the 33 panel states is present in every year.

Four independent validations were run rather than taking the release on trust. The transaction-type split (peer-to-peer, retail, utility) reproduces the aggregate total to **0.000000 per cent** across all 1,224 state-quarters, meaning two independently maintained trees within the release agree exactly. Registered users reach **45.5 crore** at 31 March 2023, against PhonePe’s own publicly stated figure of over 45 crore. Transactions total **44.3 billion** in FY2023 against roughly 84 billion for all of UPI that year (National Payments Corporation of India, 2025), consistent with the firm’s approximately 50 per cent market share. And the implied mean ticket size of Rs. 1,593 confirms amounts are denominated in rupees rather than paise.

Three caveats follow the series into the analysis and are revisited in Section 3.7. **PhonePe is one firm, not the market**, so this measures PhonePe penetration as a proxy for digital payment penetration, and because market share varies by state, cross-state levels are noisier than within-state growth — which is fortunate, since within-state growth is what the estimator uses. **Registered users are accounts, not people**, so one person with two numbers counts twice. And transactions are attributed to the **user’s registered state**, a third attribution rule alongside the branch-location and place-of-sanction rules already in the panel.

2.7 Known limitations of the data

Three defects are carried openly rather than silently repaired, and each is revisited where it bears on results.

A break in RBI’s own deposit-account series (D7). The savings-account column of BSR-2 is discontinuous for three states: Delhi runs 35.5, 38.7, **106.1**, 130.9, 159.4 million accounts, breaking from 2023 and persisting; Haryana runs 47.7, 52.6, **102.1**, breaking from 2024; Chandigarh runs 2.98, **5.03**, 3.25, spiking in 2023 and reverting. This is an error in RBI’s published source, not in extraction — the component columns reproduce the published total exactly wherever both are printed — and the likely cause is a bank reclassifying savings accounts to a head-office location. The symptom is visible downstream: deposit accounts per adult reaches 7.99 for Delhi in 2023 and 7.69 for Chandigarh in 2023, against roughly 1 to 3.5 elsewhere. Two of the four affected cells fall inside FY2018–FY2023. The decision, taken *before* estimation, was to retain them in the main specification and re-run without them as a robustness check; Section 3.4.2 reports what happened, and it is the most consequential result in the report. Goa sits at 5.45–5.73 across *all* six years and Chandigarh at 5.1–5.4 outside 2023 — high but flat, so genuine levels rather than defects.

Adult population is interpolated, not published (D6). The projections publish annual totals for 2011–2036 but age structure only at five-year anchors, so there is no published

annual adult count. The 18+ series is *derived*: the adult share is linearly interpolated between anchors and applied to the published annual total. Totals are official; adult counts are ours. Of the 266 state-years in the underlying file, 243 rest on an interpolated share. The consequence goes beyond ordinary measurement error, as Section 3.3 develops: being near-linear, the series is unusually smooth, and as the denominator of every per-adult ratio it contributes almost no independent variation — it deflates a trend without adding movement to it. A further caveat attaches to a few units: the source excludes Goa from its age-structure exercise and reports the seven smaller north-eastern states only in combination, so proxy shares were assigned and flagged.

Universe and basis of the credit-account series (D8). The two halves of *Access* initially rested on different bank universes: deposit accounts from BSR-2 *include* regional rural banks, while the quarterly BSR-1 credit series *excludes* them and records loans by place of sanction rather than utilisation. Excluding RRBs removes a median of roughly 18 per cent of credit accounts nationally, but unevenly — 65 per cent in Mizoram, 39 in Tripura, 33 in Uttar Pradesh — and the share drifts within states over time, so state fixed effects do not absorb it. Critically, it bites hardest in exactly the rural, lower-income states the hypothesis is about. The series was therefore switched to an RRB-inclusive, place-of-utilisation file covering 2010–2023. The two could not be reconciled arithmetically — differencing them produced impossible negative values for Maharashtra, because they differ on *where* a loan is counted as well as on *which* banks — so this is a replacement, not an adjustment. The cost is FY2024, which fixes FY2018–FY2023 at six years; the incidental benefit is that the last missing cell, Gujarat’s FY2023–24 income, falls outside the window, leaving the panel perfectly balanced. One asymmetry remains: deposit accounts are recorded by branch location and credit accounts by place of utilisation, and neither corresponds to depositor or borrower residence.

2.8 What the hypothesis chapters draw from this panel

Chapter 3 uses all nine series: the eight banking and demographic series across all three tracks, and the digital payment series for Track 3.

To be completed: If Hypotheses 2 and 3 use series beyond the eight above, add them to Table 2.2 and document any extra harmonisation here, so this chapter remains the single place the panel is described.

Chapter 3

Hypothesis 1: Infrastructure and the Access–Usage Gap

The hypothesis is tested on three panels, built in the order they appear here. Tracks 1 and 2 measure infrastructure physically, as branches and ATMs; both fail, and Section 3.3 shows they were always going to. Track 3 substitutes the infrastructure that actually moved over the sample window — digital payments — and recovers the pattern the hypothesis predicts. The distinction between what was pre-committed and what came afterwards is load-bearing throughout, and Section 3.6.2 states it plainly rather than leaving it to a footnote.

Table 3.1: The three tracks

Track	Window	N	Infrastructure measured	Outcome
1	FY2018–FY2023	198	Branches, ATMs	Fails; cannot be tested
2	FY2012–FY2023	384	Branches only	Fails; fragility removed
3	FY2019–FY2023	165	Branches, ATMs, UPI	Predicted pattern recovered

3.1 Hypothesis and pre-commitment

Hypothesis 1

Greater financial infrastructure raises financial inclusion, but its effect on **access** is stronger than its effect on **usage**.

Rationale. Opening an account is a supply-side decision that banks can be pushed to make. Using an account is a demand-side decision shaped by income, trust, and informality. Infrastructure removes the first constraint, not the second.

Expected: $\beta_{\text{access}} > \beta_{\text{usage}} > 0$, with the difference statistically significant.

The hypothesis is directional twice over: both effects should be *positive*, and the access effect should be *larger*. Positive and equal effects would reject it while remaining a substantive result — genuine deepening rather than formal expansion. Both effects at zero would be uninformative until the identifying variation had been examined. These readings were tabulated in advance, and Section 3.6 returns to that table.

Before any coefficient was estimated, a rule was fixed in writing for what would count as support.

Decision rule, fixed before any result was seen (D10)

Hypothesis 1 is **fully supported only** if $\beta_{\text{access}} > \beta_{\text{usage}} > 0$ holds for branches *and* for ATMs. If it holds for only one, that is reported as **partial support**, naming which. This is pre-committed; it is not to be revised after the coefficients are seen.

Two features of that rule matter for everything below. It names branches and ATMs specifically — UPI is not in it, because UPI was not in the design when it was written. And it was not revised: the physical verdict reported in Section 3.6 is the verdict the rule returns.

A second pre-commitment was recorded after the pre-estimation diagnostic failed and before estimation proceeded: a null result does **not** license the conclusion “infrastructure does not raise access”, being indistinguishable from “the regressors moved too little to detect an effect”; any null must be reported with the within-state variation figures attached (D11).

3.2 Variables and specification

3.2.1 Independent variables

Infrastructure enters in natural units, with no composite index:

$$\text{Branch}_{it} = \frac{\text{branches}_{it}}{\text{adults}_{it}} \times 10^5, \quad \text{ATM}_{it} = \frac{\text{ATMs}_{it}}{\text{adults}_{it}} \times 10^5, \quad \text{UPI}_{it} = \frac{\text{registered users}_{it}}{\text{adults}_{it}}.$$

An earlier design combined branches, ATMs and PMJDY accounts into one normalised index. It was abandoned for three reasons (D3, D10). **PMJDY had to go, and its removal proved load-bearing:** PMJDY accounts are an *account count*, so placing them among the regressors while deposit and credit account counts sit among the outcomes would put an account count on both sides of the equation, manufacturing exactly the mechanical correlation the design warns against. **Separating branches from ATMs is statistically safe:** their raw correlation is 0.904, but that is almost entirely cross-sectional, and after state and year demeaning — the variation the model actually uses — it falls to 0.305, with a variance inflation factor of 1.10. **Separating them makes a prediction an index cannot:** a branch is where an account is *opened*, an ATM where it is *used*. Abandoning the index also removed two arbitrary choices, the 0–1 rescaling of the regressors and the branches-versus-ATMs weighting, and left coefficients in interpretable units.

For Track 3, the digital measure is **registered UPI users per adult**, in levels. Three candidates existed. Transactions per capita and value per capita were rejected as regressors because they are themselves *usage* of the digital channel, and using a flow to explain a flow of the same kind would be close to circular. Registered users is an adoption stock — whether the channel is available and taken up at all — which is the role branches and ATMs play on the physical side. Levels rather than logs keeps all three infrastructure regressors in the same kind of unit (D16).

3.2.2 Dependent variables

Four indicators form two constructs, with denominators the design distinguishes deliberately: **access is measured per adult, usage per capita.**

Table 3.2: Construction of the dependent variables

Construct	Indicator	Denominator	Entangled with UPI?
<i>Access</i>	Deposit accounts per adult	Population 18+	Yes (see §3.5.2)
	Credit accounts per adult	Population 18+	No
<i>Usage</i>	Deposits per capita (Rs.)	Total population	No
	Credit per capita (Rs.)	Total population	No

Each indicator is rescaled by pooled min–max normalisation, $\tilde{x}_{it} = (x_{it} - \min x) / (\max x - \min x)$, with the extremes taken over all state-years rather than within each year, so that genuine growth over time — which is what the fixed-effects model estimates from — is preserved rather than erased. The constructs are the simple average of their two components, and the gap is their difference: $Access = \frac{1}{2}(\tilde{d} + \tilde{c})$, $Usage = \frac{1}{2}(\tilde{D} + \tilde{C})$, $AUG = Access - Usage$. The normalisation for Track 3 is held at the Track 1 scaling, so the dependent variables are identical across tracks and the results stay comparable (D15).

3.2.3 Specification

Three equations are estimated by two-way fixed effects with standard errors clustered by state. Writing $\mathbf{X}_{it}$ for the infrastructure regressors — branches and ATMs in Tracks 1 and 3, branches alone in Track 2, with UPI added in Track 3:

$$Access_{it} = \alpha_i + \lambda_t + \beta' \mathbf{X}_{it} + \gamma \ln NSDP_{it} + \varepsilon_{it} \quad (3.1)$$

$$Usage_{it} = \alpha_i + \lambda_t + \delta' \mathbf{X}_{it} + \gamma \ln NSDP_{it} + \varepsilon_{it} \quad (3.2)$$

$$AUG_{it} = \alpha_i + \lambda_t + \theta' \mathbf{X}_{it} + \gamma \ln NSDP_{it} + \varepsilon_{it} \quad (3.3)$$

State effects α_i absorb everything fixed within a state — geography, banking legacy, 2011 baselines — and year effects λ_t absorb national shocks common to all states, which over this window include COVID-19, PMJDY account opening, and the national growth of UPI. Estimates are therefore identified from **within-state variation over time, relative to the national trend in that year**; all cross-sectional variation is discarded. That is the right specification for the causal question and, as Section 3.3 shows, the source of the study’s central difficulty. With 33 clusters, cluster-robust standard errors sit at the lower end of the range in which the asymptotic approximation is reliable (Cameron and Miller, 2015).

Equation (3.3) is not a third finding. Since AUG is defined as $Access - Usage$ and all three equations share a right-hand side, linearity gives $\theta_k = \beta_k - \delta_k$ exactly. It contains no information beyond the first two equations; its purpose is to supply the **standard error on the difference**, which is the formal test of the hypothesis. The claim is not merely that $\beta > \delta$ in point estimates, but that the gap is distinguishable from zero. The identity is asserted numerically at every estimation.

3.3 The diagnostic that governs everything

The design specifies this diagnostic first, and specifies it as a *gate* rather than as description: if infrastructure barely moves within states, fixed effects have nothing to estimate from. Thresholds were fixed **before the numbers were computed** — a within-state share of variation below 10 per cent is a FAIL, below 25 per cent WEAK, at or above 25 per cent a PASS.

Table 3.3: Variance decomposition. Thresholds fixed before computation: FAIL < 10%, WEAK < 25%, PASS ≥ 25%

Variable	Overall SD	Between SD	Within SD	Within share	Verdict
<i>Physical regressors, FY2018–FY2023</i>					
Branches per lakh adults	11.455	11.574	0.805	7.0%	FAIL
ATMs per lakh adults	18.925	19.097	1.651	8.7%	FAIL
<i>Outcomes and control, same panel</i>					
<i>Access</i>	0.172	0.167	0.049	28.4%	—
<i>Usage</i>	0.200	0.201	0.029	14.4%	—
<i>AUG</i>	0.128	0.127	0.029	22.9%	—
ln(NSDP per capita)	0.541	0.528	0.147	27.1%	—
<i>Digital regressors, FY2019–FY2023 (Track 3)</i>					
UPI users per adult	—	—	—	67.1%	PASS
UPI transactions per capita	—	—	—	79.9%	PASS
UPI value per capita	—	—	—	78.8%	PASS

The gate failed for physical infrastructure — and only for physical infrastructure

Roughly **93 per cent** of the variation in branches and ATMs is *between* states, which is exactly what two-way fixed effects discard. The digital series clear the PASS threshold by a factor of three to eight.

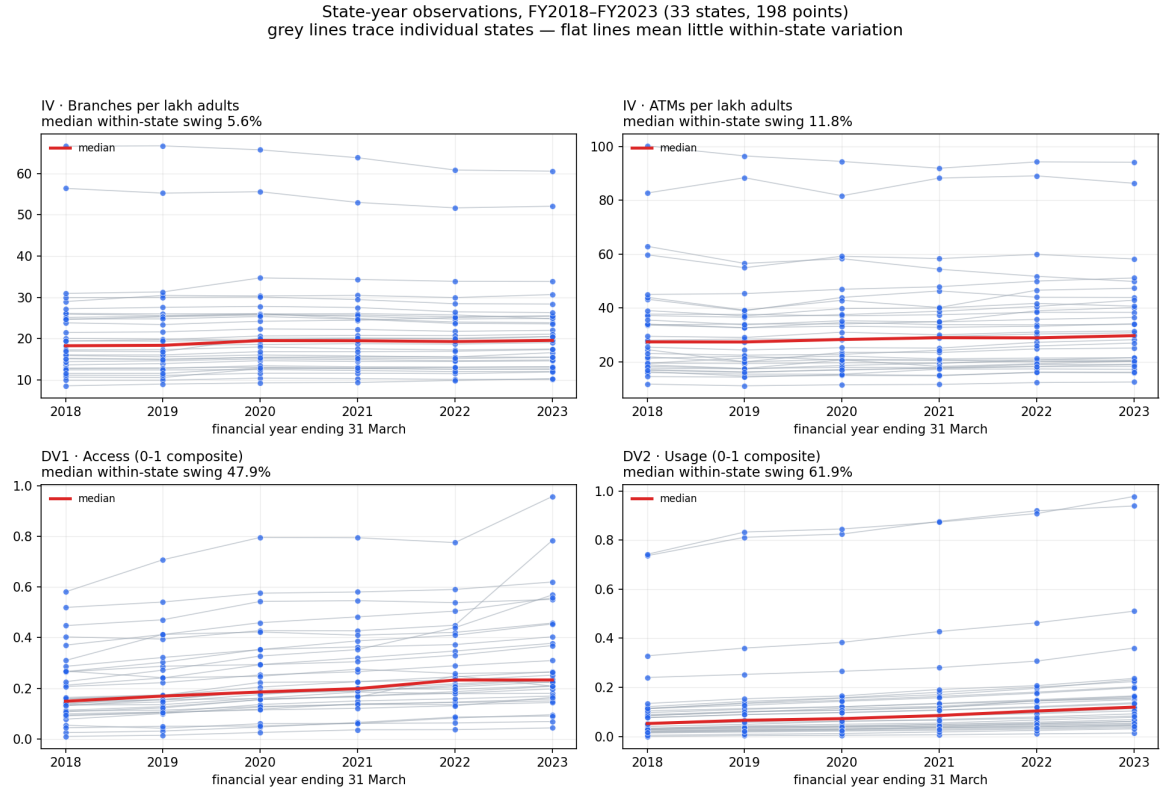


Figure 3.1: The identification problem. Each grey line traces one state across FY2018–FY2023; the red line is the cross-state median. **Top row, the regressors:** state lines run flat, with median within-state swings of 5.6 per cent (branches) and 11.8 per cent (ATMs). **Bottom row, the outcomes:** the same states climb steadily, swinging 47.9 and 61.9 per cent. Fixed effects use only the vertical movement of individual lines, and there is very little of it in the top row.

3.3.1 Why the physical variation is so thin

Three mechanisms compound, diagnosed before any coefficient was examined. **Infrastructure stocks are slow-moving:** raw branch counts swing a median of only 10.7 per cent within a state across six years — 3.6 per cent in Chandigarh, 3.7 in Punjab. **The denominator cancels much of what remains:** adult population grows a median 10.2 per cent within states over the window, in the *same direction* as branches, so branches per lakh adults swing only 5.6 per cent; because the adult series is interpolated (Section 2.7) it is almost perfectly smooth and contributes no independent variation, merely deflating a trend. **The outcomes move five to ten times more than the regressors:** credit accounts have a 30.0 per cent within share against 7.0 for branches. Wide standard errors are guaranteed by construction, whether or not an effect exists.

The problem is confined to the regressors — the outcomes and the control have ample within-state variation. The panel is not too short in general; it is too short *for physical infrastructure*. Lengthening the window helps but does not cure: the within share for branches rises from 7.0 per cent over six years to 13.4 over ten, 20.1 over twelve, and 23.2 over thirteen, still short of the PASS line. ATMs cannot follow at all, since the state-wise release begins in 2018. This is what motivates Track 2, and ultimately what motivates Track 3.

3.4 Tracks 1 and 2: physical infrastructure

Table 3.4: Two-way fixed-effects estimates, physical infrastructure. State and year effects throughout; standard errors clustered by state. Track 2 drops ATMs, which cannot be extended before 2018.

Equation	Regressor	Track 1 — FY2018–FY2023, $N = 198$			Track 2 — FY2012–FY2023, $N = 384$		
		Coef.	SE	p	Coef.	SE	p
(1) <i>Access</i>	Branches	−0.00897	0.00699	0.201	−0.00483	0.00645	0.455
	ATMs	−0.00164	0.00377	0.664	—	—	—
(2) <i>Usage</i>	Branches	−0.01264	0.00363	0.001	−0.01032	0.00570	0.071
	ATMs	−0.00109	0.00263	0.678	—	—	—
(3) <i>AUG</i>	Branches	+0.00367	0.00558	0.512	+0.00549	0.00385	0.155
	ATMs	−0.00055	0.00159	0.731	—	—	—

Within- R^2 — Track 1: 0.069, 0.042, 0.111. Track 2: −0.172, −0.478, 0.215.
Joint test (Track 1, Branch = ATM = 0): *Access* $p = 0.374$; *Usage* $p = 0.0005$; *AUG* $p = 0.710$.

The identity $\theta = \beta - \delta$ holds exactly throughout: for Track 1 branches, $-0.00897 - (-0.01264) = +0.00367$. Track 2's negative within- R^2 values are reported rather than suppressed; with two-way fixed effects the measure can go negative, and here it indicates the regressors add nothing over the fixed effects alone.

Read before interpretation: **nothing is significant in the *Access* equation** on either track, and every coefficient there is *negative*. **Exactly one coefficient is significant — branches on *Usage*, and it is negative.** ***AUG* is not significant** for any regressor on either track, so the formal test of the hypothesis returns nothing. **ATMs are insignificant everywhere.** Explanatory power is very low throughout, exactly as the gate forecast.

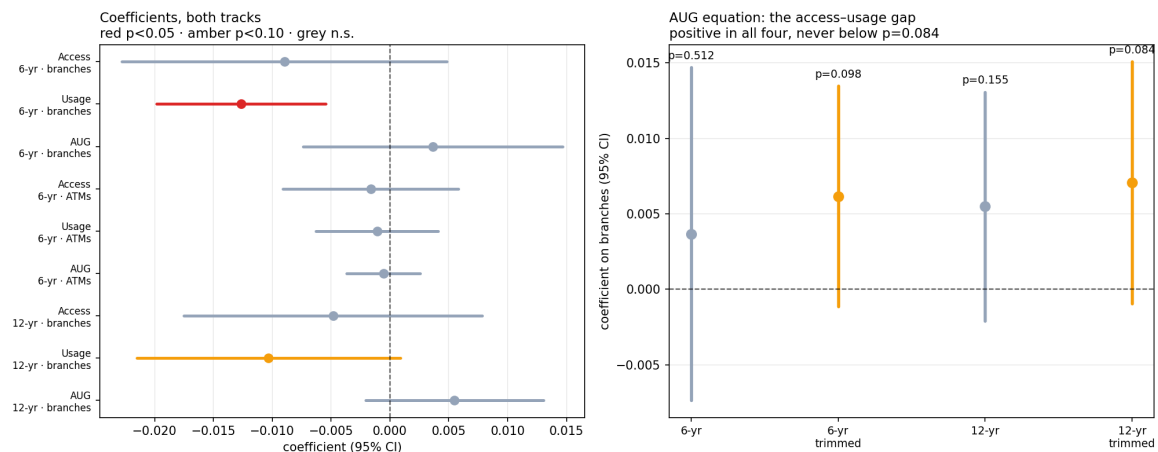


Figure 3.2: Left: every physical-infrastructure coefficient across both tracks, with 95 per cent confidence intervals. Exactly one — branches on *Usage* in Track 1 — is significant at the 5 per cent level, and it is negative. **Right:** the coefficient on branches in the *AUG* equation across all four physical specifications. It is positive in every one, in the direction the hypothesis predicts, but never attains conventional significance: p runs 0.512, 0.098, 0.155, 0.084. Reporting this as a finding would mean loosening the pre-committed rule after seeing the coefficients, so it is recorded as a suggestive pattern only.

3.4.1 Random effects is not an escape

Random effects would use both within- and between-state variation, and so appears to solve the problem. The Hausman test asks whether doing so is consistent, under the null that state effects are uncorrelated with the regressors (Hausman, 1978). On Track 1 the null is rejected decisively in all three equations ($\chi^2 = 108.3, 73.2$ and 12.4 ; $p < 0.0001, < 0.0001$ and 0.006). State effects *are* correlated with branches and ATMs — the “wealthier states attract more infrastructure” channel, confirmed rather than assumed. On Track 2 the evidence is weaker: *Access* now favours random effects ($p = 0.164$), while *Usage* and *AUG* still favour fixed effects.

The bind

The Hausman test says fixed effects are **required**. The variance decomposition says fixed effects have almost **nothing to work with**. Dropping the state fixed effects to recover precision is not available — the data itself rejects that specification. The design cannot be rescued by changing estimator.

3.4.2 Two observations decide the ATM result

All equations were re-estimated dropping the two known-defective observations identified in Section 2.7 — Delhi 2023 and Chandigarh 2023, two cells out of 198.

Table 3.5: Track 1 estimates against estimates excluding Delhi 2023 and Chandigarh 2023

Equation	Regressor	Main ($N = 198$)		Trimmed ($N = 196$)		Change
		Coef.	p	Coef.	p	
<i>Access</i>	Branches	-0.00897	0.201	-0.00570	0.078	
<i>Access</i>	ATMs	-0.00164	0.664	+0.00331	0.004	sign flip; becomes significant
<i>Usage</i>	Branches	-0.01264	0.001	-0.01185	0.000	stable
<i>Usage</i>	ATMs	-0.00109	0.678	+0.00011	0.957	sign flip
<i>AUG</i>	Branches	+0.00367	0.512	+0.00615	0.098	
<i>AUG</i>	ATMs	-0.00055	0.731	+0.00319	0.148	sign flip

Two observations out of 198 reverse the sign of every ATM coefficient

They also move the estimated effect of ATMs on *Access* from null ($p = 0.664$) to significant and positive ($p = 0.004$). A result that depends on two state-years is not a result: **nothing about ATMs can be concluded in either direction.**

This is what the gate predicted. When a regressor carries only 7 to 9 per cent within-state variation, individual observations acquire enormous leverage. The two dropped cells are moreover *known defects in RBI's published source*, not outliers chosen for being inconvenient — so the headline ATM result had been riding on data already established as wrong. On Track 2, with the window doubled and within-state variation at 15.3 per cent, the same trim produces **no sign flips anywhere**: the fragility disappears with more data, though the verdict does not change.

3.5 Track 3: substituting the infrastructure that moved

3.5.1 Why

Over FY2018–FY2023, physical banking infrastructure in India was flat while retail payments moved onto phones. If financial infrastructure was doing anything over this window, it was not doing it through branches. Figure 3.3 states that premise in numbers.

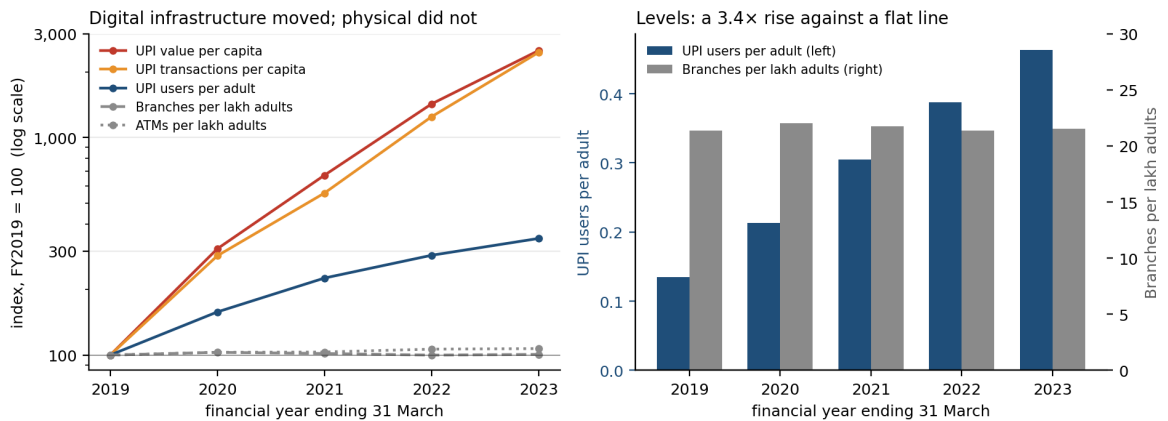


Figure 3.3: Left: unweighted state means indexed to FY2019, on a log scale. UPI value and transactions per capita rise roughly twenty-five-fold in five years; branches and ATMs per lakh adults are flat. **Right:** the same in levels — UPI users per adult rise from 0.135 to 0.464 while branches per lakh adults sit between 21.4 and 22.0 throughout.

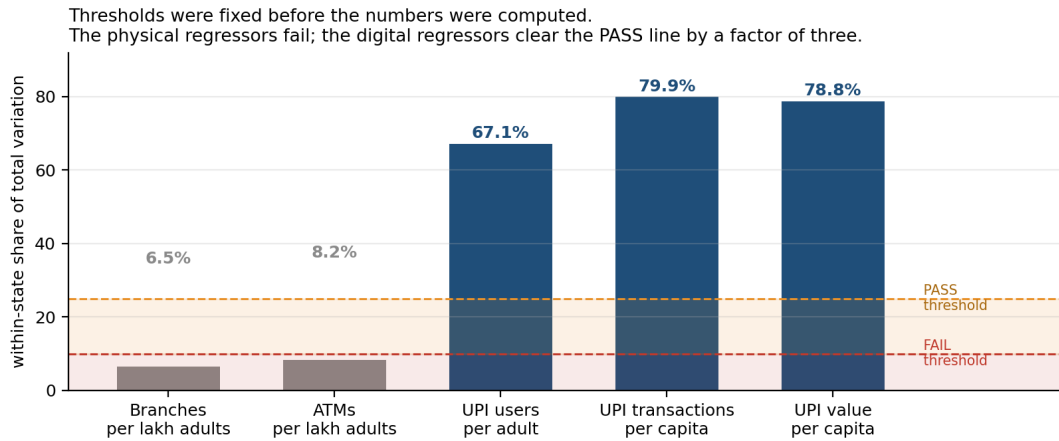


Figure 3.4: Within-state share of total variation, against thresholds fixed before the numbers were computed. The reason Tracks 1 and 2 could not test the hypothesis does not apply to Track 3.

The consequence for estimation is immediate. Within-state variation in UPI users per adult is **67.1 per cent**, against 6.5 for branches and 8.2 for ATMs on the same sample. Explanatory power rises correspondingly: within- R^2 goes from 0.069 to **0.609** for *Access* and from 0.042 to **0.863** for *Usage*. Fixed effects plus branches and ATMs explained essentially nothing; one digital variable explains most of it.

3.5.2 The test that mattered

One objection has to be dealt with before any digital coefficient can be read. PhonePe registration requires a KYC-linked bank account, so UPI adoption and *deposit-account counts* are entangled by accounting rather than by economics — the same trap that dropping PMJDY was meant to avoid, and deposit accounts are half of the *Access* construct. Credit accounts are not entangled the same way: a bank loan account is neither required for nor created by PhonePe registration.

The outcome was therefore decomposed into its four underlying indicators, with the reading fixed before the regressions were run (D17): if UPI $\rightarrow$ credit accounts is large and significant, the *Access* result is not merely definitional; if credit is null while deposits are large, the *Access* coefficient *is* the accounting relationship and *Access* must be reported credit-only thereafter; if both are large, only the credit result gets a substantive reading.

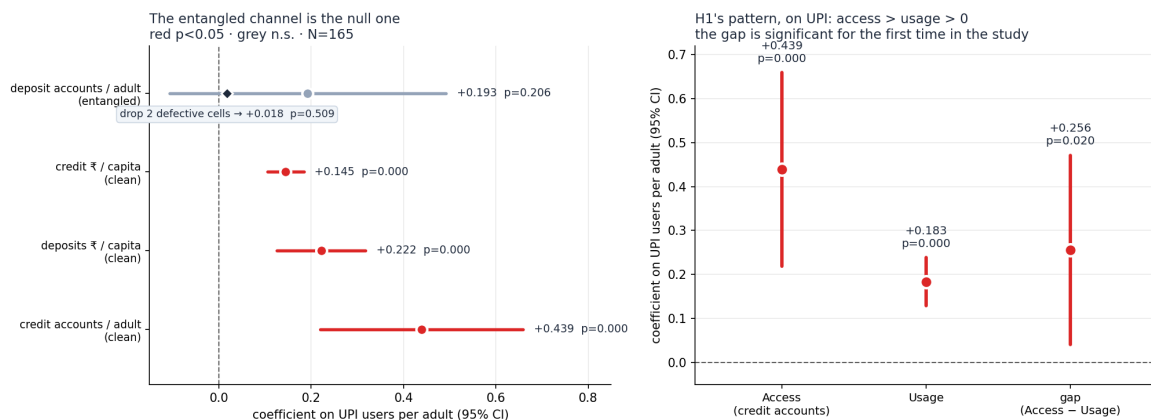


Figure 3.5: Left: the coefficient on UPI users per adult for each of the four outcome components. The entangled channel is the null one, and collapses further when the two defective cells are dropped. **Right:** the hypothesis pattern on the credit-only specification.

Table 3.6: Component decomposition: coefficient on UPI users per adult, $N = 165$, and the same estimate excluding Delhi 2023 and Chandigarh 2023

Component	Status	Coef.	p	Coef. (trimmed)	p
Deposit accounts per adult	entangled	+0.193	0.206	+0.018	0.509
Credit accounts per adult	clean	+0.439	0.0001	+0.445	0.0003
Deposits Rs. per capita	clean	+0.222	< 0.001	—	—
Credit Rs. per capita	clean	+0.145	< 0.001	—	—
<i>Access</i> (composite)	mixed	+0.316	0.0004	+0.231	0.0005

The concern was not merely unfounded — it pointed the wrong way

The **entangled** channel is the null one: UPI $\rightarrow$ deposit accounts is +0.193 ($p = 0.206$), collapsing to +0.018 ($p = 0.509$) when the two known-defective cells are dropped, with the standard error falling from 0.151 to 0.027. The **clean** channel carries everything: UPI $\rightarrow$ credit accounts is +0.439 ($p = 0.0001$) and is completely unmoved by the same trim.

The entangled component was *diluting* a real credit effect, not manufacturing one. The composite *Access* number averages a genuine effect with a defect-driven artefact, so **Access is reported credit-only from here.**

3.5.3 The hypothesis pattern, on the credit-only specification

Table 3.7: Track 3, credit-only *Access*. Two-way fixed effects, $N = 165$, standard errors clustered by state

Regressor	<i>Access</i> (credit accounts)	<i>Usage</i>	Gap (<i>AUG</i>)
UPI users per adult	+0.439 ($p < 0.001$)	+0.183 ($p < 0.001$)	+0.256 ($p = 0.020$)
Branches per lakh adults	+0.0091 ($p = 0.085$)	−0.0062 ($p = 0.020$)	+0.0153 ($p = 0.018$)
ATMs per lakh adults	−0.0007 ($p = 0.773$)	−0.0002 ($p = 0.868$)	−0.0005 ($p = 0.884$)

Four things follow. **For UPI**, $\beta_{\text{access}} > \beta_{\text{usage}} > 0$ and the gap is significant at 5 per cent — the pattern the hypothesis predicts, and the first time in this study that the access–usage gap is distinguishable from zero. **Branches turn positive on access for the first time**, once *Access* is credit-only and digital adoption is controlled for, though only at $p = 0.085$. **ATMs are null on everything**, and the ATM coefficient that looked significant under trimming in Track 1 is revealed as a composite-outcome artefact. **Physical infrastructure still fails the pre-committed rule**, and that verdict never changed.

3.5.4 Robustness

The headline coefficient was refit 33 times, dropping each state in turn.

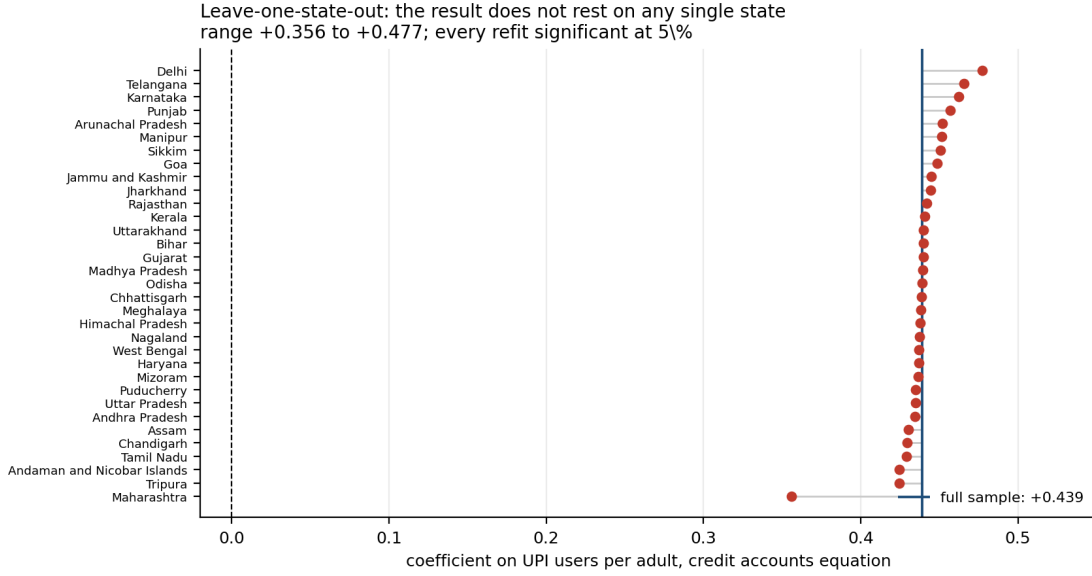


Figure 3.6: Leave-one-state-out estimates of the coefficient on UPI users per adult in the credit accounts equation. The estimate stays between +0.356 and +0.477, and no refit loses significance at the 5 per cent level. Maharashtra is the most influential single state, and the result is *stronger* without it.

The contrast with Track 1 is the point: there, two cells out of 198 flipped every ATM sign; here, removing an entire state moves the coefficient by at most 19 per cent and never changes the conclusion. The result is also unmoved by the D7 data defects, which affect savings accounts and therefore cannot touch a credit-only outcome.

3.5.5 A mechanism that fits

Digital payment history feeding credit underwriting — cash-flow-based lending, and the account aggregator framework — is a documented development in India over exactly this window, and the wider literature finds digital payment adoption spreading through network effects rather than infrastructure stock (Crouzet et al., 2023; D’Silva et al., 2019). It predicts precisely what is observed: UPI moving loan accounts while leaving deposit accounts alone. That the data is consistent with a mechanism is suggestive of it, not proof.

3.6 Verdict

3.6.1 Against the pre-committed rule

The rule requires $\beta_{\text{access}} > \beta_{\text{usage}} > 0$ for branches *and* ATMs. For branches on Track 1, the ordering holds ($-0.00897 > -0.01264$) but both coefficients are negative, so the positivity condition fails; the difference is insignificant ($p = 0.512$). For ATMs the ordering does not hold, nothing is significant, and every sign reverses on a two-cell trim. Track 2 reproduces the branch result with twice the power and none of the fragility.

Verdict on the pre-committed hypothesis

Hypothesis 1 is not supported for physical infrastructure. This is not partial support: the positivity condition fails for branches, and the ordering condition fails for ATMs. The same conclusion is reached on all three tracks, and it never changed.

Mapping onto the outcomes tabulated in advance, the row that applies is *both effects* ≈ 0 — *check within-state variation before concluding anything*. The diagnostic was run first, as instructed, and returned 7.0 and 8.7 per cent. The correct conclusion is therefore **not** that infrastructure fails to raise financial inclusion, but that **Tracks 1 and 2 cannot test the question**: the regressors barely move, the estimator the Hausman test requires discards the variation that does exist, and the estimates are unstable to two observations. The null is uninformative, exactly as fixed in advance that it would be.

3.6.2 What the digital result is, and what it is not

The digital result is exploratory by construction

The pre-committed rule named **branches and ATMs**, and it failed. **UPI was added after seeing that null**. The digital finding therefore cannot be reported as a hypothesis that was tested and passed — it is a hypothesis **generated** by this data, for a future design to test. Stating it any other way is exactly the retro-fitting that pre-committing was meant to prevent.

This is the first thing a careful reader will test, so it is stated here rather than buried. What can be said is bounded but not trivial: on a measure of financial infrastructure that genuinely moved within states, the access–usage gap widens significantly, the effect runs entirely through the channel that is not definitionally entangled with the regressor, and the result survives the removal of any single state. The identification problem that defeated Tracks 1 and 2 is solved. The causal problem is not — fixed effects remove fixed state differences and the common national trend, but not state-specific time-varying confounders, and smartphone penetration, fintech entry and youth share all rose alongside both UPI adoption and credit accounts over these five years.

3.6.3 The persistent puzzle

One physical coefficient is significant and robust across specifications: **branches on Usage, negative** ($p = 0.001$ on Track 1, 0.071 on Track 2, 0.020 on Track 3). Within a state, years with more branches per lakh adults show lower *Usage*. It is the opposite sign to anything the hypothesis anticipated, and it is treated as a puzzle rather than a result: the variation problem applies to it as much as to the nulls, *Usage* is severely compressed by the pooled normalisation, and a negative within-state relationship is at least as consistent with branches being *placed* where deposit growth is weak as with branches causing it. Nothing here identifies direction.

3.7 Limitations

Ordered by how much damage each does to the conclusions.

1. Identification — the binding limitation. On the physical tracks the regressors barely move: 7.0 and 8.7 per cent within-state variation, below the FAIL threshold fixed before looking. The null on branches and ATMs **cannot be distinguished from “not enough variation to detect anything”**, and any such null must be reported with these numbers attached. On the digital track variation is ample and that objection does not apply — but fixed effects do not remove state-specific time-varying confounders, so the digital result is a within-state correlation between two co-trending series and **no causal claim is available**. Reverse causality is live rather than hypothetical: the Hausman test rejecting random effects is direct evidence that state effects are correlated with the regressors.

2. The digital result is exploratory by construction. UPI was added after seeing the physical null (Section 3.6.2).

3. Known data defects. The deposit-account breaks in RBI’s source (Delhi 2023, Chandigarh 2023 inside the window) were enough to move the deposit-account coefficient from +0.193 to +0.018. Adult population is interpolated rather than published for 243 of 266 state-years, and being near-linear it *suppresses* within-state variation in the per-adult regressors, making the first limitation worse than it looks. Two units are dropped for want of income data, and boundary changes forced merges, so nothing state-specific can be said about either half of a merged unit.

4. Measurement. Attribution differs three ways: bank deposits and credit are recorded at branch location, loan accounts by place of sanction, and PhonePe by the user’s registered state — three different definitions of “where”, merged into one panel. The first inflates Maharashtra and Delhi, where lending is booked. **PhonePe is one firm**, roughly 45–50 per cent of UPI, with market share varying by state, so cross-state levels are noisier than within-state growth; NPCI’s total UPI series would fix this. **Usage is severely right-skewed after normalisation** — 85 per cent of observations below 0.20 — so β_{access} and β_{usage} were never on fully equivalent scales, a direct threat to the comparison itself, left unfixed deliberately in order to test the specification as written. And the study is **state-level, not person-level**: nothing here can show that *individuals* hold unused accounts, which is the mechanism the hypothesis describes.

5. Scope. The physical panel stops at FY2023, FY2024 having been traded away for a consistent bank universe; the digital panel is five years, since PhonePe data begin in 2018; ATMs cannot extend before 2018; and dropping PMJDY, though necessary, left the physical measure with nothing digital in it until PhonePe was added.

What survives all of this

The **physical null**, reported as uninformative rather than as evidence of no effect. **Branches** → **Usage negative** ($p \leq 0.020$ across three specifications and two samples), robust but with no causal reading available. **UPI** → **credit accounts positive** (+0.439), robust to dropping any single state and to the known data defects — but correlational. And **UPI** → **deposit accounts** ≈ 0 , which is the asymmetry the whole digital finding rests on.

Chapter 4

Hypothesis 2: [Title to be supplied]

4.1 Hypothesis and pre-commitment

Hypothesis 2

[Statement to be supplied.]

Rationale. [The mechanism, not the expectation.] Expected: [The sign pattern that would constitute support.]

To be completed: Fix the decision rule **before** estimating, as D10 does in Section 3.1: what counts as full support, what counts as partial support, and that it will not be revised after the coefficients are seen. Log it in Appendix A with its own D-number.

4.2 Variables and specification

To be completed: Define the independent, dependent and control variables with units and denominators, then state the estimating equation(s), fixed-effects structure, and clustering. Chapter 2 documents the shared panel — reference it rather than repeating it, and add any series not in Table 2.2.

4.3 Descriptives and diagnostics

To be completed: Means, standard deviations and ranges, flagging anything bearing on the specification — skew, compression, extreme values — **before** presenting estimates. Then run the within-state variance decomposition first, with thresholds fixed before the numbers are computed (Section 3.3), and report the verdict whether or not it is favourable.

4.4 Results and robustness

To be completed: Main estimates, reported before they are interpreted. Then re-estimate excluding the known-defective cells of Section 2.7, plus any perturbation specific to this hypothesis, reporting sign and significance changes explicitly.

4.5 Verdict, implications and limitations

To be completed: Apply the pre-committed rule as written; state whether the hypothesis is supported, partially supported or not supported, and what the result does **not** license concluding. Then substantive, policy and methodological implications, carrying the cross-cutting ones to Chapter 6; then limitations, referencing rather than restating the shared data limitations of Section 2.7.

Chapter 5

Hypothesis 3: [Title to be supplied]

5.1 Hypothesis and pre-commitment

Hypothesis 3

[Statement to be supplied.]

Rationale. [The mechanism, not the expectation.] Expected: [The sign pattern that would constitute support.]

To be completed: Fix the decision rule **before** estimating, as D10 does in Section 3.1: what counts as full support, what counts as partial support, and that it will not be revised after the coefficients are seen. Log it in Appendix A with its own D-number.

5.2 Variables and specification

To be completed: Define the independent, dependent and control variables with units and denominators, then state the estimating equation(s), fixed-effects structure, and clustering. Chapter 2 documents the shared panel — reference it rather than repeating it, and add any series not in Table 2.2.

5.3 Descriptives and diagnostics

To be completed: Means, standard deviations and ranges, flagging anything bearing on the specification — skew, compression, extreme values — **before** presenting estimates. Then run the within-state variance decomposition first, with thresholds fixed before the numbers are computed (Section 3.3), and report the verdict whether or not it is favourable.

5.4 Results and robustness

To be completed: Main estimates, reported before they are interpreted. Then re-estimate excluding the known-defective cells of Section 2.7, plus any perturbation specific to this hypothesis, reporting sign and significance changes explicitly.

5.5 Verdict, implications and limitations

To be completed: Apply the pre-committed rule as written; state whether the hypothesis is supported, partially supported or not supported, and what the result does **not** license concluding. Then substantive, policy and methodological implications, carrying the cross-cutting ones to Chapter 6; then limitations, referencing rather than restating the shared data limitations of Section 2.7.

Chapter 6

Discussion and Conclusion

6.1 Synthesis

Hypothesis 1. The claim was that financial infrastructure raises access more than usage — that expansion is formal rather than functional. Measured physically, as branches and ATMs, it is **not supported**, and the more consequential result is why. Only 7.0 per cent of the variation in branches per lakh adults, and 8.7 per cent of that in ATMs, occurs within states; roughly 93 per cent is cross-sectional, which is precisely what fixed effects discard. The Hausman test then rejects random effects, closing off the specification that would have used that variation, and two defective observations out of 198 prove enough to reverse the sign of every ATM coefficient. Doubling the window to twelve years doubles the usable variation and removes the fragility, but leaves the verdict unchanged.

Measured digitally, the picture reverses. UPI adoption varies 67.1 per cent within states over the same window, explanatory power rises roughly ninefold, and the predicted pattern appears: *Access* + 0.439, *Usage* + 0.183, gap +0.256 at $p = 0.020$. The effect runs entirely through credit accounts and not at all through deposit accounts — the opposite of what an accounting artefact would produce, since deposit accounts are the component entangled with UPI registration by construction. The asymmetry is the finding. But UPI entered the design after the physical null was seen, so this is a hypothesis generated by the data rather than one tested and passed.

To be completed: Summarise the verdicts on Hypotheses 2 and 3 here in the same form — what was claimed, what was found, what the result does not license — then add a short paragraph on whether the three point in a common direction. If they share the panel of Chapter 2, note explicitly whether the within-state variation problem of Section 3.3 affects them too: it is a property of the data, not of Hypothesis 1.

6.2 Implications for policy

Measure usage, not ownership. Account ownership is near saturation, and the marginal account opened now is very likely to be dormant: roughly 35 per cent of Indian adults with an account hold an inactive one (Demirgüç-Kunt et al., 2022), and about 26 per cent of PMJDY accounts are reported inoperative (Press Information Bureau, Government of India, 2025). The RBI’s Financial Inclusion Index already weights usage (45 per cent) above access (35 per cent) (Reserve Bank of India, 2025a), and the evidence here supports that ordering. Targets expressed in accounts opened measure the dimension that is no longer binding.

The binding infrastructure has changed, and measurement should follow. Branches per lakh adults move a median of 5.6 per cent within a state over six years; UPI adoption moved by orders of magnitude over the same period. Whatever the long-run effect of physical

premises, they are not the margin on which annual variation in financial inclusion was determined over FY2019–FY2023. A framework that monitors branch and ATM density while treating digital payment adoption as background is measuring the stationary part of the system.

If the credit channel is real, it is a policy object. The digital result is correlational, but the association it identifies — digital payment adoption moving *loan* accounts while leaving deposit accounts untouched — is consistent with payment history entering credit underwriting, which is a deliberate feature of Indian financial architecture over this period. That makes it a testable and governable channel rather than an incidental one, and it sharpens what an evaluation should look for.

Data quality is itself a policy variable. A discontinuity in one column of one RBI return, affecting three states, was enough to reverse the sign of every ATM coefficient in Track 1 and to inflate a null deposit-account coefficient tenfold in Track 3. Series used for programme evaluation warrant the revision and documentation discipline applied to headline aggregates.

6.3 Implications for method

Diagnose the identifying variation before estimating. The variance decomposition was specified as a gate, with thresholds fixed before the numbers were computed, and it correctly forecast the instability the robustness checks later demonstrated. Had coefficients been examined first, the null on *Access* would have read as a substantive finding about infrastructure, and the significant ATM coefficient under trimming as a discovery. Both readings would have been wrong, and neither obviously so. The same statistic then did positive work: it is what identified digital adoption as a regressor capable of answering the question, rather than merely another variable to try. A wide confidence interval is ambiguous between “there is no effect” and “there is no variation from which to detect one”, and this distinction carries the whole report.

Decompose a composite outcome when any component is entangled with the regressor. The headline *Access* coefficient on UPI would have been reportable as it stood. Splitting it revealed that the entangled component was null and the clean component carried everything — so the composite was averaging a real effect with a defect-driven artefact, and the naive concern about mechanical correlation pointed the wrong way. The rule for reading that decomposition was fixed before it was run, which is what made the outcome interpretable either way.

Pre-registration is most valuable when results are ambiguous. Had the estimates been clean, the rule fixed in advance would have been redundant. They were not. It is also what forces the honest statement of the digital result’s status: the rule named branches and ATMs, so no reading of the UPI finding can claim the hypothesis was tested and passed.

Slow-moving regressors and short panels are a recurring trap. Infrastructure, institutions and capital stocks all move slowly; two-way fixed effects is the default estimator; and the Hausman test will typically forbid retreat to random effects in exactly the settings where between-state variation is most correlated with the outcome. The reconsideration of Burgess and Pande (2005) by Buliskeria and Baxa (2022) makes the same point from the other direction. The remedy applied here — find the infrastructure that actually moved — is more generally available than it looks.

6.4 Directions for further work

Three things would strengthen this, in order of value. **An identification strategy for UPI** is the first and largest: the current digital result is a within-state correlation between two co-trending series, and candidate instruments include state-level 4G rollout timing or the staggered geography of Jio’s 2016–17 entry. **Total UPI rather than one firm** would remove the varying-market-share problem, since NPCI publishes state-level aggregates that would replace the PhonePe proxy. **Extending the digital panel** would take it from five years to eight, the PhonePe series running to FY2026 while the banking series stop at FY2023. Beyond these, sub-state disaggregation would raise precision substantially, and three measurement problems are tractable: replacing the interpolated adult denominator with Census-based age shares, estimating on logs or ranks rather than pooled min–max levels to address the compression of *Usage*, and reconciling the three attribution rules currently merged into one panel.

6.5 Conclusion

This report asked whether the expansion of financial infrastructure in India has been formal rather than functional — whether it opens accounts faster than it puts them to use. The question is well motivated: India has effectively solved account ownership while recording the highest rate of account dormancy in the world.

Asked of *physical* infrastructure, the question cannot be answered with these data. Hypothesis 1 is not supported, but the null is uninformative: branches and ATMs do not move enough within Indian states over six years, or even twelve, for a fixed-effects panel to detect their effect, and the specification test that would license using cross-sectional variation instead rejects that estimator decisively. Two observations out of 198 reverse the sign of every ATM coefficient, which demonstrates the point rather than merely asserting it.

Substituting the infrastructure that actually moved changes the picture. UPI adoption varies abundantly within states, and on that measure the predicted pattern appears: access rises more than usage, and the gap between them is significantly different from zero for the first time in the study. It does so entirely through credit accounts rather than deposit accounts — an asymmetry that is the reverse of what a mechanical artefact would generate, and one that fits a documented channel, digital payment history feeding credit underwriting.

The honest headline

A state panel cannot identify the effect of *physical* banking infrastructure on financial inclusion, because branches and ATMs barely move within states. Substituting the infrastructure that actually moved — digital payments — the picture reverses: UPI adoption predicts access more strongly than usage, significantly so, and it does it entirely through credit accounts rather than deposit accounts. **The identification problem is solved; the causal one is not.**

Two boundaries must travel with that. The pre-committed hypothesis named branches and ATMs and it failed; UPI was added afterwards, so the digital finding is a hypothesis this data *generated* and not one it tested. And fixed effects remove fixed state differences and the common national trend, not state-specific time-varying confounders — smartphone penetration, fintech entry and youth share all rose alongside both UPI adoption and credit accounts.

What remains is a clear direction rather than a closed question. The variation this design needed and did not have exists in abundance on the digital side; what it now lacks is exogeneity, not power. That is a more tractable problem than the one it started with, and the discipline

that made this report's negative result legible — fixing the decision rule and the diagnostic thresholds before seeing the numbers — is what should be carried into solving it.

Appendix A

Decision Log

Every non-obvious choice taken during data preparation and estimation was recorded when it was taken, with its rationale and consequences, in an append-only log, reproduced here in full. Its purpose is auditability: each choice can be verified as made for a stated reason and — for the decisions constraining interpretation — as made *before* the results it constrains were seen.

Table A.1: Decision log, D1–D17

#	Decision	Rationale and consequence
D1	Drop Lakshadweep and Dadra & Nagar Haveli and Daman & Diu	No per-capita NSDP published for either in any year, so neither can carry the income control. 35 units become 33; both are very small.
D2	Merge J&K with Ladakh, and DNH with Daman and Diu, in every year	Boundaries changed mid-sample and fixed effects require each unit to mean the same thing throughout. Sources record the change in different years, so summing all components in all years is the only rule that works for every source at once.
D3	Drop PMJDY; infrastructure is branches and ATMs only	Removes the only digital component, so the measure is <i>physical banking infrastructure</i> . Later found load-bearing rather than cosmetic: PMJDY accounts are an account count, and retaining them would have put an account count on both sides of the regression.
D4	Keep Gujarat; leave its FY2023–24 income cell missing	Not sourceable — Gujarat is absent from every Handbook income table that year. Superseded by D8, which moves the cell outside the window.
D5	Year alignment: banking $Y \equiv \text{income}$ $(Y-1)-Y \equiv \text{population}$ 1 March Y	All stocks at 31 March; the easiest place to introduce a silent one-year offset. Implemented as a canonical column in D9.
D6	Disclose that adult population is interpolated	Age structure is published only every five years; 243 of 266 state-years use an estimated 18+ share applied to published annual totals.
D7	Keep Delhi, Haryana and Chandigarh despite defective deposit-account values	The savings-account column jumps discontinuously (Delhi 39m to 106m in 2023; Haryana 53m to 102m in 2024; Chandigarh spikes in 2023 and reverts). Extraction is correct — the break is in RBI’s source. Retain in the main specification, then re-run without them as a robustness check.
D8	Switch loan accounts to the RRB-inclusive file; panel becomes FY2018–FY2023	Fixes a bank-universe mismatch so every variable counts the same banks. The files could not be merged arithmetically — they also differ on <i>where</i> a loan is counted, and differencing gave impossible negative values for Maharashtra. Cost: FY2024. Benefit: removes the Gujarat gap, leaving 198 balanced rows.

continued on next page

Table A.1 continued from previous page

#	Decision	Rationale and consequence
D9	Add one canonical year column, <code>fy_end</code> , to all eight files	<code>fy_end</code> is the calendar year of the 31 March a row refers to; merge, filter and take the year fixed effect on it. Only the income file needed converting; native labels retained so the original coding stays auditable. Not named plain <code>year</code> , because a bare 2018 is exactly the ambiguity that causes the bug.
D10	No composite index; branches and ATMs enter as two separate regressors. Final	Raw correlation 0.904 is almost entirely cross-sectional; after demeaning it falls to 0.305, VIF 1.10. Removes two arbitrary choices and puts coefficients in real units. Fixes the pre-committed decision rule: full support only if $\beta_{\text{access}} > \beta_{\text{usage}} > 0$ for branches <i>and</i> ATMs; one only is partial support, naming which; not to be revised after the coefficients are seen.
D11	Proceed to estimation despite the diagnostic gate failing	Within-state shares of 7.0 and 8.7 per cent fall below the 10 per cent FAIL threshold fixed in advance. Proceeding was directed explicitly. Consequence fixed now so it cannot be revised later: a null does not license the conclusion “infrastructure does not raise access”, and every null must be reported with the 7.0 and 8.7 per cent figures attached. A significant result, by contrast, would be strong. The known skew in <i>Usage</i> is <i>not</i> repaired, so the specification tested is the one originally written.
D12	Extended panel FY2012–FY2023, branches only. Revises D10	The six-year panel could not test the hypothesis and was unstable to two observations; extending the window is the only remedy in these data. The binding constraint is per-capita income, published from 2011–12. ATMs are dropped (the release begins in 2018), so the branches-versus-ATMs comparison is no longer possible; no composite index is reintroduced. Deposit accounts are spliced at 2018, verified exact at the overlap (1,911,503,615 from both sources). The six-year results are not discarded — this is an additional specification, not a replacement.
D13	Merge Andhra Pradesh and Telangana in all years. Extends D2; twelve-year panel only	Telangana shows zero branches, deposits and credit for 2012–2014, having been carved out of Andhra Pradesh in June 2014. Branches confirm it: AP falls 9,900 to 6,290 in 2015 as Telangana appears with 4,721, the combined series running continuously 9,900 to 11,011. Panel becomes $32 \times 12 = 384$; the six-year results are unaffected. Had this been missed, three state-years would have entered with zero infrastructure and zero deposits — an outlier positioned precisely to manufacture a spurious positive coefficient.
D14	New direction: add PhonePe Pulse digital-payment data	The physical-infrastructure results came back null or negative. Working interpretation: branches and ATMs are the wrong infrastructure for this period, since digital payments boomed over exactly this window while the physical footprint was flat to declining. Nothing estimated is discarded; the contrast between the tracks is the point. Caveat: PhonePe is one firm, roughly 45–50 per cent of UPI volume, so a proxy for digital-payment penetration. How it enters is an open decision, to be logged before anything is estimated.
D15	Build the merged digital panel: 33 states $\times$ FY2019–FY2023, 165 rows	FY2018 is dropped, since the first complete financial year of digital data is FY2019. Denominators follow the existing rule: registered users (a stock) per adult; transactions and value (flows) per capita. <i>Access</i> , <i>Usage</i> and <i>AUG</i> are <i>not</i> recomputed on the 165 rows — they keep the scaling pooled over the original 198, so the dependent variables are identical across tracks and the results stay comparable. Every banking column is asserted identical to the six-year panel. Left open deliberately: whether the digital series enters as a regressor, a second outcome, or an interaction — recorded so the choice is made on reasoning rather than on whichever specification returns stars.

continued on next page

Table A.1 continued from previous page

#	Decision	Rationale and consequence
D16	Resolves D15: UPI enters as a third regressor , measured as registered users per adult, in levels	Three candidate measures existed. Transactions and value per capita were rejected because they are themselves <i>usage</i> of the digital channel, so using them to predict the <i>Usage</i> equation would be close to circular — a flow explaining a flow of the same kind. Registered users is an adoption stock, the role branches and ATMs play on the physical side. Levels rather than logs keeps all three infrastructure regressors in the same kind of unit. Model, diagnostics and clustering are otherwise identical to the earlier specification. Note: because the sample is one year shorter, the branches and ATMs coefficients here are not directly comparable to those from the six-year panel — same regressors, different sample.
D17	Decompose <i>Access</i> and <i>Usage</i> into their four indicators and re-run, to test whether the UPI result is an accounting relationship	PhonePe registration requires a KYC-linked bank account, so UPI adoption and deposit-account counts — half of <i>Access</i> — are definitionally entangled: the same problem dropping PMJDY was meant to remove. Credit accounts are not entangled the same way, a loan account being neither required for nor created by PhonePe registration. Rule fixed before running: if UPI → credit accounts is large and significant, the <i>Access</i> result is not merely definitional; if credit is null while deposits are large, the <i>Access</i> coefficient <i>is</i> the accounting relationship and <i>Access</i> must thereafter be reported credit-only; if both are large, only the credit result gets a substantive reading, with no averaging of the two conclusions. Outcome: the rule returned “credit-only survives”. The entangled channel is null, the clean channel carries the whole effect and is unmoved by the data-defect check. <i>Access</i> is reported credit-only from there — not because the rule forced it, but because the defect check showed the composite was averaging a real credit effect with an artefact.

Every decision above was recorded when it was taken, before the result it constrains was known. Three are load-bearing for how the findings must be read: **D10** fixes the criterion for Hypothesis 1 and names branches and ATMs, which is why the digital result cannot be reported as a test that was passed; **D11** fixes in advance that a null on physical infrastructure is uninformative rather than evidence of no effect; and **D17** fixes how the component decomposition is to be read, before it was run.

To be completed: Append decisions taken for Hypotheses 2 and 3 with the next available D-numbers, so a single log covers the whole report.

Acknowledgement of AI Assistance

In the interest of transparency, the authors record the following use of an AI assistant in the preparation of this report.

Claude, an AI assistant developed by Anthropic, was used through the Claude Code command-line interface ([Anthropic, 2026](#)) at three points in this project:

1. **Coding.** Assistance with writing and debugging the Python code used to retrieve, parse, and reconcile the source data — in particular the RBI Handbook HTML parsers, the ATM workbook parser, the DBIE/CIMS retrieval client, and the population-projection PDF extraction — together with the panel-construction, variable-construction, estimation, diagnostic, and plotting scripts.
2. **Formalising the hypothesis and the decision rule.** Assistance with translating the hypothesis into the three-equation specification in Section 3.2.3, with articulating the pre-committed decision rule recorded as D10, and with identifying the consequences that had to be fixed in advance of estimation (D11).
3. **Drafting the report.** Assistance with structuring and drafting the text of this document.

All data retrieval was from the official sources cited in Chapter 2, and all figures reported here were produced by the analysis scripts and checked against independently published aggregates as described in Section 2.5. The choice of hypotheses, the interpretation of the results, the decision to report the negative finding as the study’s headline, and the conclusions drawn are the authors’ own, and the authors take full responsibility for the contents of this report.

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